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One-Page Report: National Energy Scenario (India)

Introduction:

India's energy sector is central to its economic growth, industrialization, and rising living standards. As one of the world's fastest-growing major economies, India faces the dual challenge of **meeting increasing energy demand** while ensuring **energy security, affordability, and sustainability**.

Energy Demand & Consumption:

- India is among the **top global energy consumers**, driven by population growth, urbanization, and expanding industries.
- Energy demand is rising across **power, transport, residential, and industrial sectors**.
- Per-capita energy consumption remains below the global average, indicating **future growth potential**.

Energy Mix:

India's energy supply is diversified but still **fossil-fuel dominant**:

- **Coal:** Primary source for electricity generation
- **Oil:** Crucial for transport and imports-dependent
- **Natural Gas:** Growing but limited share
- **Renewables:** Rapidly expanding (solar, wind, hydro, biomass)
- **Nuclear:** Small but strategic contribution

Coal continues to support base-load power, while renewables are transforming the generation landscape.

Renewable Energy Growth:

India has made significant progress in clean energy:

- Major investments in **solar and wind capacity**
- Large-scale solar parks and rooftop programs
- Push toward **green hydrogen and energy storage**

Objectives include reducing carbon intensity and achieving long-term climate targets.

Energy Security Challenges:

- Heavy dependence on **crude oil imports**
- Exposure to **global price volatility**
- Need for **domestic resource development**
- Grid integration of intermittent renewables

Policy & Government Initiatives:

Key focus areas:

- Expansion of **renewable energy capacity**
- Promotion of **electric vehicles (EVs)**
- Energy efficiency through programs like LED adoption
- Strengthening **transmission & smart grids**
- Emphasis on **energy access and rural electrification**

Environmental Considerations:

- Fossil fuel use contributes to **emissions and air pollution**
- Transition toward **low-carbon technologies** is essential
- Increased adoption of **clean fuels and efficiency measures**

Emerging Trends:

- Digitalization & **smart energy management**
- Growth of **distributed generation**
- Battery storage & hybrid renewable systems
- Decarbonization of transport and industry

Conclusion:

India's national energy scenario reflects a **transition phase**: balancing traditional energy sources with an accelerating shift toward **renewables and sustainability**. Continued investment, policy support, and technological innovation will be critical to achieving **energy security, economic development, and climate goals**.